GlassBox: Personalized, Privacy-Preserving, and Explainable Multimodal Risk Assessment for Online Exam Proctoring

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Abstract

The rapid expansion of online examinations has increased reliance on automated proctoring, yet fixed thresholds, isolated events, and opaque decisions can confuse legitimate individual differences with potentially unusual behavior while providing limited evidence for instructor review. GlassBox addresses this limitation by shifting proctoring from uniform surveillance toward individualized, evidence-based risk assessment. We present a browser-based framework integrating personalized behavioral calibration, multimodal temporal reasoning, interpretable machine learning, privacy-preserving processing, and empirical robustness evaluation. During an initial calibration period, GlassBox establishes an individual baseline for gaze and head-pose behavior, allowing subsequent observations to be interpreted as deviations from the student's own behavioral norm. Real-time visual signals from MediaPipe Face Landmarker are combined with browser evidence, including tab visibility, focus changes, and clipboard activity, and organized into temporal behavioral episodes capturing persistence, recurrence, and cross-signal concurrence rather than isolated events. The resulting features are processed by an interpretable logistic-regression classifier that produces probabilistic risk estimates and exposes feature-level contributions underlying each assessment. For visual robustness and fairness analysis, an open-source RetinaFace detector is evaluated independently of the real-time MediaPipe pipeline using demographically diverse facial data to quantify detection performance and subgroup disparities. The completed framework is evaluated through student- and session-independent validation, fixed-threshold and rule-based baselines, personalized-versus-global calibration comparisons, modality and component ablations, predictive-performance and probability-calibration metrics, and demographic robustness analysis. GlassBox follows a privacy-by-design, human-in-the-loop approach, performing sensitive facial processing locally wherever feasible and treating automated predictions as decision support rather than proof of misconduct. By transforming opaque, one-size-fits-all monitoring into individualized, traceable, and auditable evidence-based assessment, GlassBox provides a practical foundation for more transparent, accountable, and scientifically testable online examination proctoring.

Keywords: Online Exam Proctoring, Personalized Calibration, Explainable AI, Multimodal Risk Assessment, Privacy-Preserving AI, Algorithmic Fairness, Behavioral Risk Assessment, Human-in-the-Loop AI

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